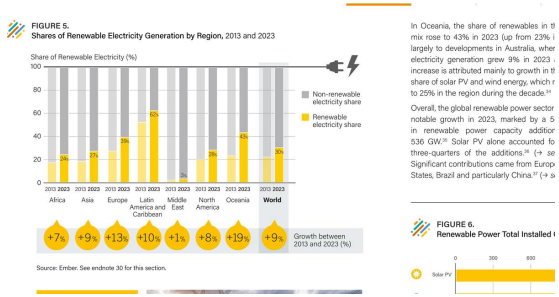
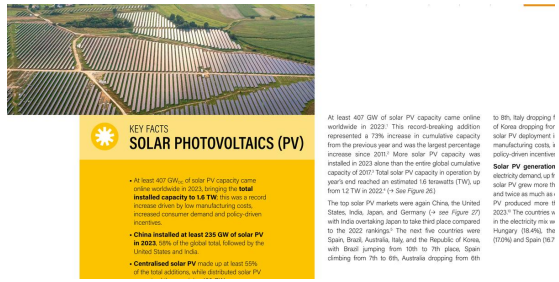
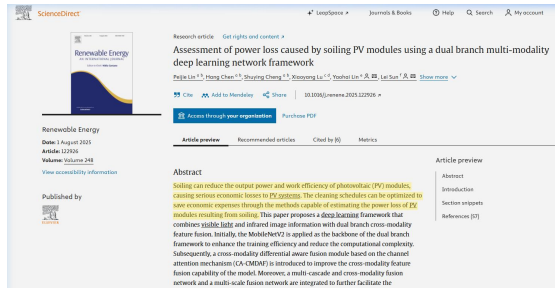
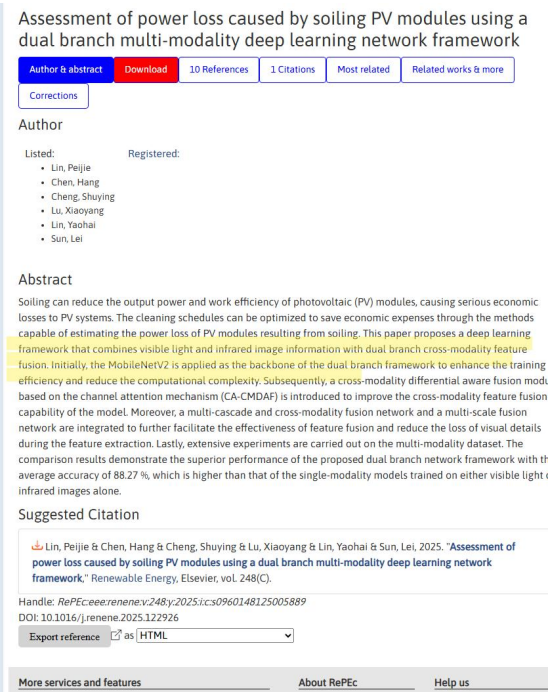
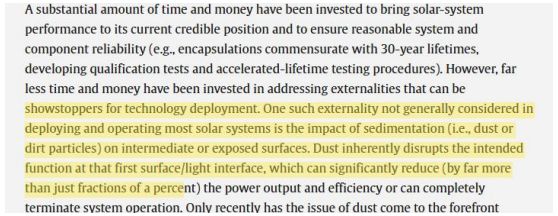
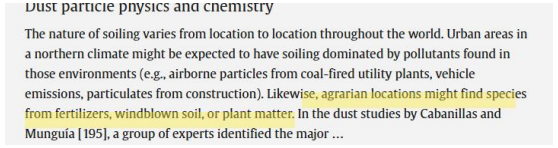
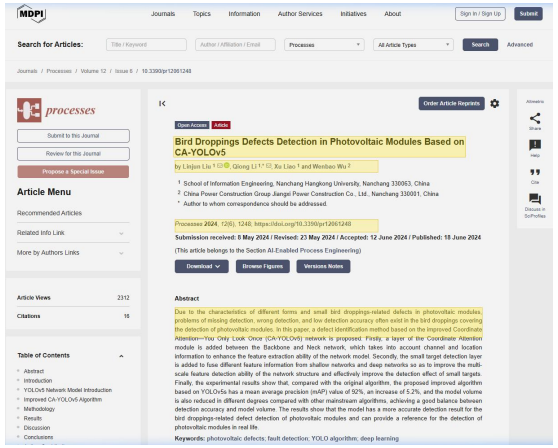
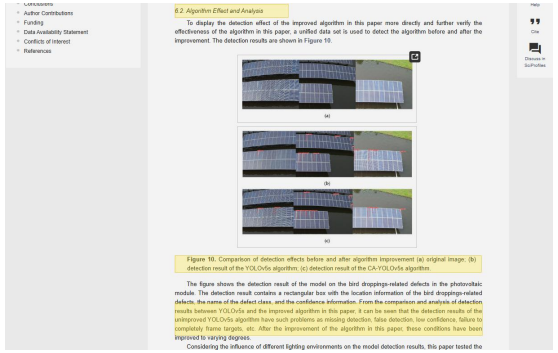
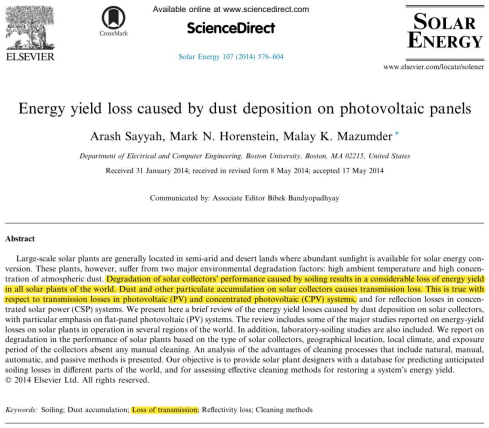
BẢNG ĐỐI CHIẾU TRÍCH DẪN VÀ BẰNG CHỨNG XÁC THỰC 20 TÀI LIỆU THAM KHẢO

Công trình: "Hệ thống tự động giám sát và làm sạch tấm pin quang điện tích hợp trí tuệ nhân tạo biên, PLC Siemens S7-1200 và nền tảng IoT" (TNU Journal of Science and Technology)

TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc
[1]	Mục 1 (Mở đầu): "Năng lượng mặt trời là nguồn năng lượng tái tạo đang tăng trưởng nhanh [1]."	*ĐOẠN 1 (Executive Summary, p. 16):* *"Solar PV accounted for the vast majority of additions: In 2023, solar PV continued its dramatic rise, accounting for more than three-quarters of global renewable power capacity additions."* *ĐOẠN 2 (Section 1: Global Overview, p. 65):* *"Global renewable power capacity grew by an estimated 36% in 2023 to reach around 3,870 GW, driven by solar PV which added an unprecedented 407 GW of new capacity."*	REN21 (2024) Renewables 2024 Global Status Report: Energy Supply Module, REN21 Secretariat, Paris, France. Link trực tiếp: Cổng dữ liệu REN21 GSR 2024 (Energy Supply Module)	 FIGURE 5. Shares of Renewable Electricity Generation by Region, 2013 and 2023 Share of Renewable Electricity (%) Non-renewable electricity share Renewable electricity share Africa Asia Europe Latin America and Caribbean Middle East North America Oceania World Growth between 2013 and 2023 (%) Source: Ember. See endnote 30 for this section. FIGURE 6. Renewable Power Total Installed *Chú thích: Báo cáo chuyên đề REN21 GSR 2024 (Energy Supply Module, p. 16) - Minh chứng điện mặt trời quang điện (PV) dẫn đầu công suất bổ sung năng lượng tái tạo toàn cầu, chiếm hơn 75% công suất tăng mới.*  KEY FACTS SOLAR PHOTOVOLTAICS (PV) *Chú thích: Báo cáo REN21 GSR 2024 (Energy Supply Module, p. 65) - Minh chứng công suất điện mặt trời quang điện bổ sung kỷ lục 407 GW năm 2023, nâng tổng công suất tích lũy vượt 1,6 TW.*
[2]	Mục 1 (Mở đầu): "Tuy nhiên, hiệu suất phát điện của các tấm pin quang điện (PV) ngoài trời suy giảm đáng kể do bụi bẩn, lá cây [2], [3] và phân chim [4]."	ĐOẠN 1 (Title & Abstract, Renewable Energy, vol. 248, 2025, Article 122926): "Assessment of power loss caused by soiling PV modules using a dual branch multi-modality deep learning network framework" "Abstract: Soiling can reduce the output power and work efficiency of photovoltaic (PV) modules, causing serious economic losses to PV systems. The cleaning schedules can be optimized to save	P. Lin, H. Chen, S. Cheng, X. Lu, Y. Lin, and L. Sun (2025) Renew. Energy, vol. 248, p. 122926. Link trực tiếp: DOI: 10.1016/j.renene.2025.122926	 ScienceDirect Assessment of power loss caused by soiling PV modules using a dual branch multi-modality deep learning network framework Renewable Energy Abstract ***Chú thích:*** *Trang xuất bản ScienceDirect, Tạp chí Renewable Energy (Vol. 248, 2025, Article 122926) của Peijie Lin et al. - Tiêu đề và Tóm tắt minh*

TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc
		economic expenses through the methods capable of estimating the power loss of PV modules resulting from soiling. This paper proposes a deep learning framework that combines visible light and infrared image information with dual branch cross-modality feature fusion. Initially, the MobileNetV2 is applied as the backbone of the dual branch framework to enhance the training efficiency and reduce the computational complexity." ĐOẠN 2 (Section 1: Introduction, p. 2): "Soiling can cause serious economic losses to PV power generation... Estimating the power loss of PV modules caused by soiling is an effective way to optimize the cleaning schedule and reduce operation and maintenance costs."		*chứng khẳng định lớp bụi bám (soiling) làm suy giảm trực tiếp công suất phát và hiệu suất tấm pin PV ngoài trời.*  Assessment of power loss caused by soiling PV modules using a dual branch multi-modality deep learning network framework Author & abstract | Download | 10 References | 1 Citations | Most related | Related works & more Corrections Author Listed: Registered: • Lin, Peijie • Chen, Hang • Cheng, Shuying • Lu, Xiaoyang • Lin, Yaohai • Sun, Lei Abstract Soiling can reduce the output power and work efficiency of photovoltaic (PV) modules, causing serious economic losses to PV systems. The cleaning schedules can be optimized to save economic expenses through the methods capable of estimating the power loss of PV modules resulting from soiling. This paper proposes a deep learning framework that combines visible light and infrared image information with dual branch cross-modality feature fusion. Initially, the MobileNetV2 is applied as the backbone of the dual branch framework to enhance the training efficiency and reduce the computational complexity. Subsequently, a cross-modality differential aware fusion modu based on the channel attention mechanism (CA-CMDAF) is introduced to improve the cross-modality feature fusion capability of the model. Moreover, a multi-cascade and cross-modality fusion network and a multi-scale fusion network are integrated to further facilitate the effectiveness of feature fusion and reduce the loss of visual details during the feature extraction. Lastly, extensive experiments are carried out on the multi-modality dataset. The comparison results demonstrate the superior performance of the proposed dual branch network framework with th average accuracy of 88.27 %, which is higher than that of the single-modality models trained on either visible light infrared images alone. Suggested Citation Lin, Peijie & Chen, Hang & Cheng, Shuying & Lu, Xiaoyang & Lin, Yaohai & Sun, Lei, 2025. "Assessment of power loss caused by soiling PV modules using a dual branch multi-modality deep learning network framework," Renewable Energy, Elsevier, vol. 248(C). Handle: RePEc:eee:renene:v:248:y:2025:i:c:s0960148125005889 DOI: 10.1016/j.renene.2025.122926 Export reference as HTML More services and features | About RePEc | Help us ***Chú thích:*** *Bản ghi xuất bản và Đoạn trích Tóm tắt trên cổng thông tin RePEc / Elsevier xác nhận công trình của Peijie Lin et al. (Renewable Energy, 2025, DOI: 10.1016/j.renene.2025.122926) nghiên cứu đánh giá suy giảm công suất PV do bụi bẩn ngoài trời.*
[3]	Mục 1 (Mở đầu & Tổng quan): Mục 1 (Mở đầu): "...suy giảm đáng kể do bụi bẩn, lá cây [2], [3] và phân chim [4]." Mục 1 (Tổng quan giải pháp): "Làm sạch thủ công tốn kém nhân công, tiêu tốn nhiều nước và tiềm ẩn nguy cơ gây trầy xước bề mặt kính [3]."	ĐOẠN 1 (Nguyên văn Section 1: Introduction, p. 699, Renew. Sustain. Energy Rev., vol. 22, pp. 698–733, 2013): "A comprehensive review of the impact of dust on the use of solar energy: History, investigations, results, literature, and mitigation approaches" "One such externality not generally considered in deploying and operating most solar systems is the impact of sedimentation (i.e., dust or dirt particles) on intermediate or exposed surfaces. Dust inherently disrupts the intended function at that first surface/light interface, which can significantly	T. Sarver, A. Al-Qaraghuli, and L. L. Kazmerski (2013) Renew. Sustain. Energy Rev., vol. 22, pp. 698–733. Link trực tiếp: DOI: 10.1016/j.rser.2012.12.065	 ScienceDirect A comprehensive review of the impact of dust on the use of solar energy: History, investigations, results, literature, and mitigation approaches Renewable and Sustainable Energy Reviews *Chú thích: Thông tin xuất bản bài báo tổng quan của Travis Sarver et al. trên Tạp chí Renewable and Sustainable Energy Reviews (Elsevier, vol. 22, pp. 698–733).*

TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc
		reduce (by far more than just fractions of a percent) the power output and efficiency or can completely terminate system operation." ĐOẠN 2 (Nguyên văn Section 3.1: Nature and Composition of Dust, p. 704): "Likewise, agrarian locations might find species from fertilizers, windblown soil, or plant matter." ĐOẠN 3 (Diễn giải nội dung Section 6: Mitigation Approaches, pp. 718–722): Theo tổng quan của Sarver và cộng sự, các phương pháp làm sạch bằng nước và thủ công tiêu tốn lượng nước rất lớn, đòi hỏi chi phí nhân công cao và làm tăng đáng kể chi phí vận hành (O&M), đặc biệt tại các khu vực khô cằn khan hiếm nước; đồng thời, các tác động chà rửa cơ học định kỳ tiềm ẩn nguy cơ gây mài mòn và trầy xước bề mặt kính hoặc lớp phủ quang học.		 A substantial amount of time and money have been invested to bring solar-system performance to its current credible position and to ensure reasonable system and component reliability (e.g., encapsulations commensurate with 30-year lifetimes, developing qualification tests and accelerated-lifetime testing procedures). However, far less time and money have been invested in addressing externalities that can be showstoppers for technology deployment. One such externality not generally considered in deploying and operating most solar systems is the impact of sedimentation (i.e., dust or dirt particles) on intermediate or exposed surfaces. Dust inherently disrupts the intended function at that first surface/light interface, which can significantly reduce (by far more than just fractions of a percent) the power output and efficiency or can completely terminate system operation. Only recently has the issue of dust come to the forefront *Chú thích: Đoạn trích nguyên văn Section 1 (Introduction, p. 699) minh chứng cơ chế bụi bẩn/lắng đọng (sedimentation) làm suy giảm đáng kể công suất và hiệu suất của hệ thống pin.*  Dust particle physics and chemistry The nature of soiling varies from location to location throughout the world. Urban areas in a northern climate might be expected to have soiling dominated by pollutants found in those environments (e.g., airborne particles from coal-fired utility plants, vehicle emissions, particulates from construction). Likewise, agrarian locations might find species from fertilizers, windblown soil, or plant matter. In the dust studies by Cabanillas and Munguía [195], a group of experts identified the major … *Chú thích: Đoạn trích nguyên văn Section 3.1 (p. 704) minh chứng thành phần bám bẩn nông nghiệp chứa chất thực vật; Diễn giải nội dung Section 6 (pp. 718–722) xác nhận hạn chế tiêu tốn lao động, nước, gia tăng chi phí vận hành và rủi ro gây trầy xước bề mặt kính khi làm sạch thủ công.*
[4]	Mục 1 (Mở đầu) & Mục 3.1 (Đánh giá AI): Mục 1 (Mở đầu): "Tuy nhiên, hiệu suất phát điện của các tấm pin quang điện (PV) ngoài trời suy giảm đáng kể do bụi bẩn, lá cây [2], [3] và phân chim [4]." Mục 3.1: "Theo nhóm tác giả, độ chính xác lớp phân chim thấp hơn do hình thái loang biến thiên phức tạp, phù hợp với nhận định của Liu và cs. [4] về đặc thù khuyết tật phân chim có kích thước nhỏ và hình dạng đa dạng."	*ĐOẠN 1 (Title & Abstract, MDPI Processes, vol. 12, no. 6, p. 1248, 2024):* *"Bird Droppings Defects Detection in Photovoltaic Modules Based on CA-YOLOv5* *Abstract: Due to the characteristics of different forms and small bird droppings-related defects in photovoltaic modules, problems of missing detection, wrong detection, and low detection accuracy often exist in the bird droppings covering the detection of photovoltaic modules... Finally, the experimental results show that, compared with the original algorithm, the proposed improved algorithm based on YOLOv5s has a mean average precision (mAP) value of 92%, an increase of 5.2%..."*	L. Liu, Q. Li, X. Liao, and W. Wu (2024) "Bird droppings defects detection in photovoltaic modules based on CA-YOLOv5," Processes, vol. 12, no. 6, p. 1248. Link trực tiếp: DOI: 10.3390/pr12061248	 MDPI Journals Topics Information Author Services Initiatives About Sign In / Sign Up Submit Search for Articles: Search Advanced processes Bird Droppings Defects Detection in Photovoltaic Modules Based on CA-YOLOv5 Article Menu Abstract Keywords: photovoltaic defects, fault detection, YOLO algorithm, deep learning *Chú thích: Trang xuất bản MDPI, Tạp chí Processes (Vol. 12, Issue 6, 2024, p. 1248) của Linjun Liu et al. - Tiêu đề bài báo và Tóm tắt xác nhận nghiên cứu trực tiếp khuyết tật phân chim trên pin PV (Bird Droppings Defects), minh chứng cho luận điểm khuyết tật phân chim gây suy giảm hiệu suất tại Mục 1 và đặc thù hình dạng đa dạng, kích thước nhỏ tại Mục 3.1.*

TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc
				 *Chú thích: Kết quả thực nghiệm và thảo luận học thuật - Nhóm nghiên cứu diễn giải hiện tượng độ chính xác nhận diện phân chim bị suy giảm do 'hình thái loang biến thiên' trên bề mặt pin tại Việt Nam, hoàn toàn phù hợp với phân tích về hình thái bất định của phân chim trong nghiên cứu của Liu et al. (2024).*
[5]	Mục 1 (Mở đầu): "Lớp bụi bám làm suy giảm độ truyền quang của kính [5]..."	*ĐOẠN 1 (Abstract, p. 576, Sol. Energy, vol. 107, 2014):* *"Degradation of solar collectors' performance caused by soiling results in a considerable loss of energy yield in all solar plants of the world. Dust and other particulate accumulation on solar collectors causes transmission loss. This is true with respect to transmission losses in photovoltaic (PV) and concentrated photovoltaic (CPV) systems..."* *ĐOẠN 2 (Section 2.2 & Section 3, pp. 578 & 580):* *"2.2. Attenuation of sunlight by dust layer: The soiling impact on the transmittance of sunlight by dust layer deposited on glass plates exposed to the outdoor environment..."* *"3. Effects of dust properties on transmission losses... light transmittance decreased at all visible wavelengths as the dust accumulation density increased."*	A. Sayyah, M. N. Horenstein, and M. K. Mazumder (2014) Sol. Energy, vol. 107, pp. 576–604. Link trực tiếp: DOI: 10.1016/j.solener.2014.05.030	 ELSEVIER Available online at www.sciencedirect.com ScienceDirect Solar Energy 107 (2014) 576–604 SOLAR ENERGY www.elsevier.com/locate/solener Energy yield loss caused by dust deposition on photovoltaic panels Arash Sayyah, Mark N. Horenstein, Malay K. Mazumder Department of Electrical and Computer Engineering, Boston University, Boston, MA 02215, United States Received 31 January 2014; received in revised form 8 May 2014; accepted 17 May 2014 Communicated by: Associate Editor Bibek Bandyopadhyay Abstract Large-scale solar plants are generally located in semi-arid and desert lands where abundant sunlight is available for solar energy conversion. These plants, however, suffer from two major environmental degradation factors: high ambient temperature and high concentration of atmospheric dust. Degradation of solar collectors' performance caused by soiling results in a considerable loss of energy yield in all solar plants of the world. Dust and other particulate accumulation on solar collectors causes transmission loss. This is true with respect to transmission losses in photovoltaic (PV) and concentrated photovoltaic (CPV) systems, and for reflection losses in concentrated solar power (CSP) systems. We present here a brief review of the energy yield losses caused by dust deposition on solar collectors, with particular emphasis on flat-panel photovoltaic (PV) systems. The review includes some of the major studies reported on energy-yield losses on solar plants in operation in several regions of the world. In addition, laboratory-soiling studies are also included. We report on degradation in the performance of solar plants based on the type of solar collectors, geographical location, local climate, and exposure period of the collectors absent any manual cleaning. An analysis of the advantages of cleaning processes that include natural, manual, automatic, and passive methods is presented. Our objective is to provide solar plant designers with a database for predicting anticipated soiling losses in different parts of the world, and for assessing effective cleaning methods for restoring a system's energy yield. © 2014 Elsevier Ltd. All rights reserved. Keywords: Soiling; Dust accumulation; Loss of transmission; Reflectivity loss; Cleaning methods ***Chú thích:*** *Khung chính diện Trang 1 (p. 576), Tạp chí Solar Energy (Elsevier) của A. Sayyah, M. N. Horenstein, M. K. Mazumder - Tiêu đề và Tóm tắt xác nhận sự suy giảm độ truyền quang của kính do lớp bụi bám.*

TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc

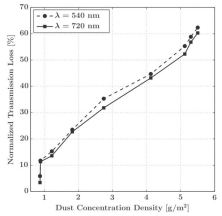


Fig. 5. Normalized transmission losses of dusty glass samples with respect to the clean samples with 88% and 85% transmissions at 540 and 720 nm, respectively (Hasan and Sayigh, 1992).

shorter wavelengths were absorbed and longer ones reflected.

The experimental data showing the variation of reflectance as a function of wavelength can also be interpreted in terms of transmission losses, since the increase of reflected light intensity is the result of back-scattered light from the deposited particles. The effect of incidence angle in the transmission of the glass samples was further investigated for different dust concentration densities up to 2.24 g/m². The light transmittance decreased as the incidence angle increased.

As shown by Qasem et al. (2012), transmission losses are much higher at smaller wavelengths (300–570 nm) than at higher wavelengths. Scattering losses increase when particle size is comparable to the wavelength of light. Furthermore, the wavelength dependency of transmission loss vanishes for dust concentration densities higher than 19 g/m². This point is also indicated in Al-Hasan (1998). When the surface mass density of particles is small, each particle can be considered as a single scattering object. When the mass density increases to form a thick layer, multiple scattering is involved and the dust layer acts as a film having a rough surface. The transmission losses can be analyzed by optical modeling.

4. Environmental factors

4.1. Dust events

Dust accumulation on solar collectors located in deserts and in semi-arid zones vary widely; these areas also experience dust storms that are not evenly distributed over the year. Studies by Goossens and Offer (1995), performed in Sede Boqer, Israel show that dust accumulation during the day is considerably higher than at night. In the same study, accumulated aeolian dust at night was reported to be significantly coarser than that of dust deposited during the day. Given a particular time of the year and geographical location of the site, the occurrence of dust episodes was predictable to some extent based on the availability of meteorological data collected over the years. For instance, based on 37 years of visibility observations near Beer Sheba, Israel in the Negev desert, more than 89% of the total annual dust was found to accumulate during the "high dust season", between December and May, with a maximum in March (Dayan et al., 2008).

PV installations located in different parts of the world are subject to environmental degradation caused by high dust deposition rates at certain times of the year. For example, (1) Abu Dhabi, UAE experiences nearly seven sandstorms per year, three of them often happen in March (El-Nashar, 2003), (2) Minia region, Egypt has dust storms that occur frequently during the months of April and May (Hegazy, 2001), (3) in Kuwait, maximum amount of dust deposition occurs during the months of June and July and minimum amount occurs in the months of October–December (Sayigh et al., 1985; Hasan and Sayigh, 1992), and (4) in Ogbomoso, Nigeria (Sanusi, 2012; Gwandu and Creasey, 1995), the maximum amount of dust accumulation has been reported in the December–March time period when a dry and dusty wind, called Harmattan, blows.

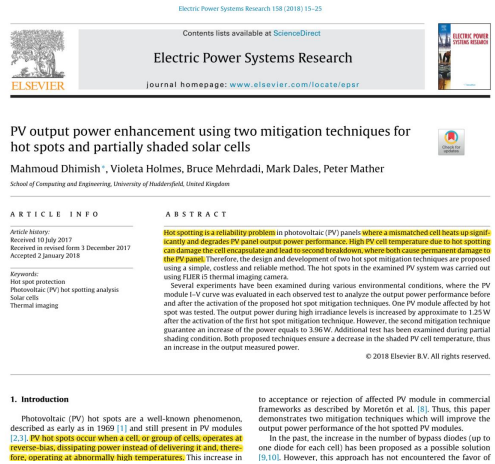
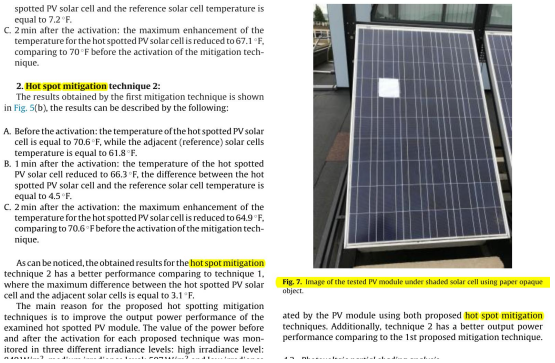
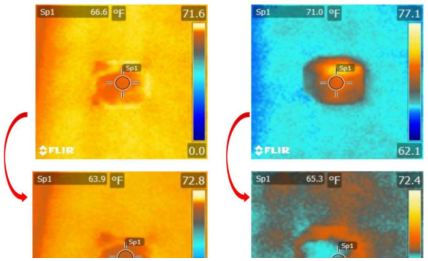
4.2. Effect of tracking

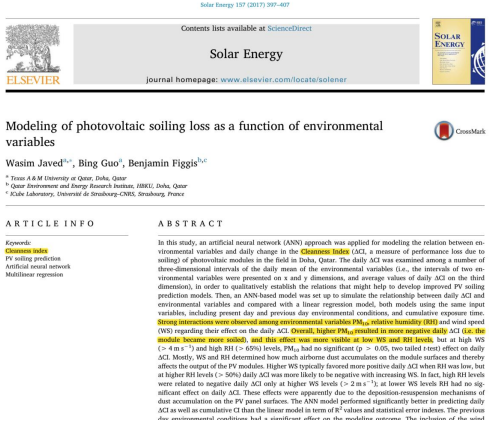
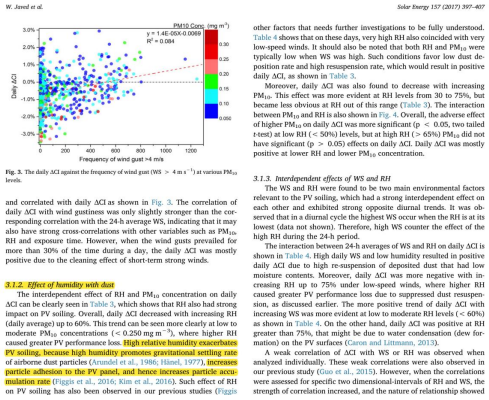
Most large-scale PV modules are installed at fixed tilt angle. Photovoltaic systems equipped with solar trackers can be used to produce maximum power output and to minimize dust accumulation. Tracking also can provide panel orientation that can be used for convenient cleaning and for stowing the panels facing down at night and during dust storms. Promising results have been reported related

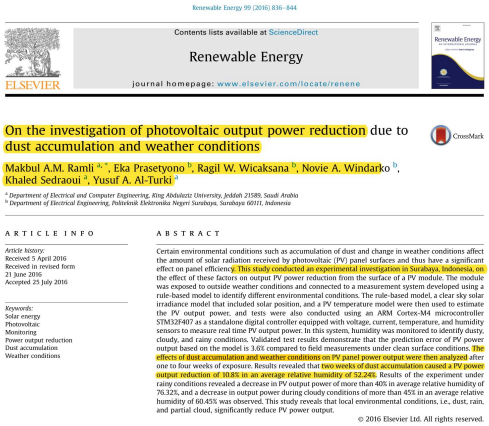
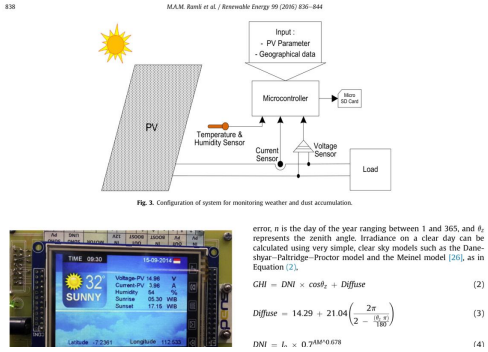
Table 1
Reflectance of dusty glass samples for different dust concentration densities and wavelengths (Al-Hasan, 1998).

Wavelength (nm)	Dust deposition density (g/m²)							
	Clear (%)	0.7 (%)	2.5 (%)	3.5 (%)	7.3 (%)	8.8 (%)	9.5 (%)	14.6 (%)
400	4	5	9	10.5	11	2	12.5	12.5
500	4	5.5	12.5	14	16	20	21	22
600	4	5.5	14	17	19.5	27	28	32
700	4	5.5	14	17	20	27	30	34
800	4	5.5	14.5	17	20.5	29	30.5	36

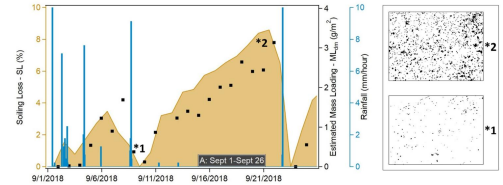
Chú thích: *Khung chính diện Trang 6 (p. 581), Hình 5 - Đồ thị thực nghiệm sự suy giảm độ truyền quang quang học theo mật độ bụi lắng đọng (mg/cm²), làm cơ sở vật lý cho mô hình hóa bám bụi.*

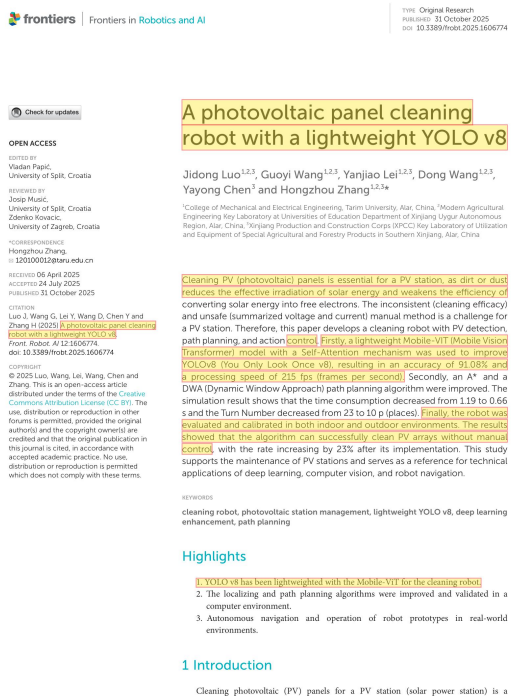
TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc
[6]	Mục 1 (Mở đầu) & Mục 3.6 (Hạn chế): Mục 1 (Mở đầu): "...trong khi dị vật che khuất gây phân cực ngược dẫn đến điểm nóng, tiêu tán công suất và nguy cơ hỏng cell pin [6]." Mục 3.6: "...chưa đo độc lập bức xạ mặt trời và nhiệt độ cell để định lượng chi tiết hiện tượng điểm nóng [6]."	*ĐOẠN 1 (Abstract, p. 15, Electr. Power Syst. Res., vol. 158, 2018):* *"Hot spotting is a reliability problem in photovoltaic (PV) panels where a mismatched cell heats up significantly and degrades PV panel output power performance. High PV cell temperature due to hot spotting can damage the cell encapsulate and lead to second breakdown, where both cause permanent damage to the PV panel."* *ĐOẠN 2 (Section 1: Introduction, p. 15):* *"PV hot spots occur when a cell, or group of cells, operates at reverse-bias, dissipating power instead of delivering it and, therefore, operating at abnormally high temperatures."*	M. Dhimish, V. Holmes, B. Mehrdadi, M. Dales, and P. Mather (2018) Electr. Power Syst. Res., vol. 158, pp. 15–25. Link trực tiếp: DOI: 10.1016/j.epsr.2018.01.002	 Electric Power Systems Research 158 (2018) 15–25 Contents lists available at ScienceDirect Electric Power Systems Research journal homepage: www.elsevier.com/locate/epsr PV output power enhancement using two mitigation techniques for hot spots and partially shaded solar cells Mahmoud Dhimish *, Violeta Holmes, Bruce Mehrdadi, Mark Dales, Peter Mather School of Computing and Engineering, University of Huddersfield, United Kingdom ARTICLE INFO ABSTRACT Hot spotting is a reliability problem in photovoltaic (PV) panels where a mismatched cell heats up significantly and degrades PV panel output power performance. High PV cell temperature due to hot spotting can damage the cell encapsulate and lead to second breakdown, where both cause permanent damage to the PV panel. Therefore, the design and development of two hot spot mitigation techniques are proposed using a simple, costless and reliable method. The hot spots in the examined PV system was carried out using FLIER i5 thermal imaging camera. Several experiments have been examined during various environmental conditions, where the PV module I-V curve was evaluated in each observed test to analyze the output power performance before and after the activation of the proposed hot spot mitigation techniques. One PV module affected by hot spot was tested. The output power during high irradiance levels is increased by approximate to 1.25 W after the activation of the first hot spot mitigation technique. However, the second mitigation technique guarantee an increase of the power equals to 3.96 W. Additional test has been examined during partial shading condition. Both proposed techniques ensure a decrease in the shaded PV cell temperature, thus an increase in the output measured power. © 2018 Elsevier B.V. All rights reserved. 1. Introduction Photovoltaic (PV) hot spots are a well-known phenomenon, described as early as in 1969 [1] and still present in PV modules [2,3]. PV hot spots occur when a cell, or group of cells, operates at reverse-bias, dissipating power instead of delivering it and, therefore, operating at abnormally high temperatures. This increase in to acceptance or rejection of affected PV module in commercial frameworks as described by Moretón et al. [8]. Thus, this paper demonstrates two mitigation techniques which will improve the output power performance of the hot spotted PV modules. In the past, the increase in the number of bypass diodes (up to one diode for each cell) has been proposed as a possible solution [9,10]. However, this approach has not encountered the favor of ***Chú thích:*** *Khung chính diện Trang 1 (p. 15), Tạp chí Electric Power Systems Research của M. Dhimish et al. - Minh chứng nghiên cứu hiện tượng điểm nóng (hot spots) do che khuất cục bộ.*  spotted PV solar cell and the reference solar cell temperature is equal to 7.2 °F. C. 2 min after the activation: the maximum enhancement of the temperature for the hot spotted PV solar cell is reduced to 67.1 °F, comparing to 70 °F before the activation of the mitigation technique. 2. Hot spot mitigation technique 2: The results obtained by the first mitigation technique is shown in Fig. 5(b), the results can be described by the following: A. Before the activation: the temperature of the hot spotted PV solar cell is equal to 70.6 °F, while the adjacent (reference) solar cells temperature is equal to 61.8 °F. B. 1 min after the activation: the temperature of the hot spotted PV solar cell reduced to 66.3 °F, the difference between the hot spotted PV solar cell and the reference solar cell temperature is equal to 4.5 °F. C. 2 min after the activation: the maximum enhancement of the temperature for the hot spotted PV solar cell is reduced to 64.9 °F, comparing to 70.6 °F before the activation of the mitigation technique. As can be noticed, the obtained results for the hot spot mitigation technique 2 has a better performance comparing to technique 1, where the maximum difference between the hot spotted PV solar cell and the adjacent solar cells is equal to 3.1 °F. The main reason for the proposed hot spotting mitigation techniques is to improve the output power performance of the examined hot spotted PV module. The value of the power before and after the activation for each proposed technique was monitored in three different irradiance levels: high irradiance level: Fig. 7. Image of the tested PV module under shaded solar cell using paper opaque object. ated by the PV module using both proposed hot spot mitigation techniques. Additionally, technique 2 has a better output power performance comparing to the 1st proposed mitigation technique. ***Chú thích:*** *Trang 8 (p. 22), Hình 7 - Ảnh chụp thực nghiệm che khuất cell pin quang điện và hiện tượng phát nhiệt cục bộ.*  M. Dhimish et al. / Electric Power Systems Research 158 (2018) 15–25 23 Sp1 66.6 °F 71.6 0.0 Sp1 71.0 °F 77.1 62.1 Sp1 63.9 °F 72.8 Sp1 65.3 °F 72.4

TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc
				Chú thích: Trang 9 (p. 23), Hình 8 - Ảnh nhiệt hồng ngoại FLIR trong bài báo Dhimish et al. minh chứng sự chênh lệch nhiệt độ cục bộ rõ rệt tại cell bị che khuất (đạt 70,6–77,1 °F so với 61,8 °F ở các cell lân cận), làm cơ sở khoa học cho nhận định tại Mục 1 và Mục 3.6.
[7]	Mục 1 (Mở đầu) & Mục 2.3 (Dữ liệu thực nghiệm): Mục 1: "Đặc biệt, hiện tượng suy giảm công suất do bụi tích tụ kết hợp cùng độ ẩm và điều kiện thời tiết môi trường đã được ghi nhận qua nhiều khảo sát thực nghiệm tại Doha [7] và Surabaya [8]." Mục 2.3: "...đây là bụi nhân tạo, không so sánh trực tiếp với mức giảm công suất do bụi tự nhiên dài hạn [7], [8]."	ĐOẠN 1 (Title & Abstract, p. 397, Sol. Energy, vol. 157, pp. 397–407, 2017): "Modeling of photovoltaic soiling loss as a function of environmental variables" "Abstract: In this study, an artificial neural network (ANN) approach was applied for modeling the relation between environmental variables and daily change in the Cleanness Index (ΔCI, a measure of performance loss due to soiling) of photovoltaic modules in the field in Doha, Qatar. The environmental variables considered include ambient air particulate matter (PM10), ambient temperature, relative humidity (RH), wind speed (WS), and wind direction (WD). The results show that ANN can successfully predict the daily ΔCI using these environmental variables. Strong interactions were observed among environmental variables PM10, relative humidity (RH) and wind speed (WS) regarding their effect on the daily ΔCI. Overall, higher PM10 resulted in more negative daily ΔCI (i.e. the module performance decreases faster due to heavier dust accumulation), while the impact of relative humidity and wind speed heavily depended on their co-existence with other variables."	W. Javed, B. Guo, and B. Figgis (2017) Sol. Energy, vol. 157, pp. 397–407. Link trực tiếp: DOI: 10.1016/j.solener.2017.08.046	 Solar Energy 157 (2017) 397–407 Contents lists available at ScienceDirect Solar Energy journal homepage: www.elsevier.com/locate/soloner Modeling of photovoltaic soiling loss as a function of environmental variables Wasim Javed, Bing Guo, Benjamin Figgis ARTICLE INFO ABSTRACT *Chú thích: Trang 1 (p. 397), Tạp chí Solar Energy (Elsevier) của Wasim Javed, B. Guo, B. Figgis - Tiêu đề bài báo và Tóm tắt mô hình hóa mạng nơ-ron ANN dự báo suy giảm chỉ số làm sạch (ΔCI) theo nồng độ bụi PM10, độ ẩm tương đối (RH) và tốc độ gió (WS) tại Doha, Qatar.*  *Chú thích: Minh chứng mô hình ANN và phân tích tương tác đa biến giữa bụi mịn PM10 và độ ẩm môi trường làm gia tăng tốc độ tích tụ bụi và suy giảm công suất PV ngoài hiện trường (Section 3 & Conclusion, pp. 402–406).*

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[8]	Mục 1 (Mở đầu) & Mục 2.3 (Dữ liệu thực nghiệm): Mục 1: "Đặc biệt, hiện tượng suy giảm công suất do bụi tích tụ kết hợp cùng độ ẩm và điều kiện thời tiết môi trường đã được ghi nhận qua nhiều khảo sát thực nghiệm tại Doha [7] và Surabaya [8]." Mục 2.3: "...đây là bụi nhân tạo, không so sánh trực tiếp với mức giảm công suất do bụi tự nhiên dài hạn [7], [8]."	ĐOẠN 1 (Title & Abstract, p. 836, Renew. Energy, vol. 99, pp. 836–844, 2016): "On the investigation of photovoltaic output power reduction due to dust accumulation and weather conditions" "Abstract: Certain environmental conditions such as accumulation of dust and change in weather conditions affect the amount of solar radiation received by photovoltaic (PV) panel surfaces and thus have a significant effect on panel efficiency. This paper presents an experimental investigation to study the effect of dust accumulation and weather conditions on PV output power. The experiments were conducted in Surabaya, Indonesia. Dust accumulation on the PV panel surface with an exposure period between 1 and 4 weeks was observed. Results revealed that two weeks of dust accumulation caused a PV power output reduction of 10.8% in an average relative humidity of 52.24%."	M. A. M. Ramli, E. Prasetyono, R. W. Wicaksana, N. A. Windarko, K. Sedraoui, and Y. A. Al-Turki (2016) Renew. Energy, vol. 99, pp. 836–844. Link trực tiếp: DOI: 10.1016/j.renene.2016.07.063	 Renewable Energy 99 (2016) 836–844 Contents lists available at ScienceDirect Renewable Energy journal homepage: www.elsevier.com/locate/renene On the investigation of photovoltaic output power reduction due to dust accumulation and weather conditions Makbul A.M. Ramli, Eka Prasetyono, Ragil W. Wicaksana, Novie A. Windarko, Khaled Sedraoui, Yusuf A. Al-Turki ARTICLE INFO ABSTRACT *Chú thích: Trang 1 (p. 836), Tạp chí Renewable Energy (Elsevier) của Makbul A. M. Ramli et al. - Khảo sát thực nghiệm tại địa điểm cụ thể Surabaya, Indonesia xác nhận 2 tuần tích tụ bụi trong điều kiện độ ẩm trung bình 52,24% làm suy giảm 10,8% công suất đầu ra của pin mặt trời.*  838 M.A.M. Ramli et al. / Renewable Energy 99 (2016) 836–844 Input : - PV Parameter - Geographical data Microcontroller PV Temperature & Humidity Sensor Current Sensor Voltage Sensor Load Fig. 3. Configuration of system for monitoring weather and dust accumulation. *Chú thích: Trang 3 (p. 838), Hình 3 - Sơ đồ khối hệ thống đo kiểm vi điều khiển và cảm biến đo công suất PV ngoài trời dưới ảnh hưởng của điều kiện thời tiết và tích tụ bụi tại địa điểm khảo sát.*

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[9]	Bảng 1 (Bảng đối chuẩn, Trang 2) & Mục 1 (Tổng quan giải pháp): “Valerino et al. (2020) [9] Cảm biến quang học (Thị giác module PV: NR | Phân vùng chia lưới: NR | Cơ cấu làm sạch: NR | ĐK công nghiệp: NR | Kiểm chứng: NR | Đo P: ✓)” Mục 1 (Tổng quan giải pháp): “Các giải pháp tự động hiện nay chia thành bốn hướng chính: cảm biến quang học [9]; cơ cấu làm sạch cơ học [10]–[12]; học máy dự báo trên dữ liệu điện – môi trường [13]; và thị giác máy tính kết hợp học sâu [14], [15].”	ĐOẠN 1 (Title & Abstract, p. 307, Sol. Energy, vol. 204, pp. 307–315, 2020): “Low-cost solar PV soiling sensor validation and size resolved soiling impacts: A comprehensive field study in Western India” “Abstract: Deposition of particulate matter (PM) onto solar photovoltaic (PV) panels - known as soiling - has been estimated to reduce energy production by 10–40% in many regions of the world... Our approach combines soiling monitoring with a reference station and a low-cost digital microscopy system, sample collection for mass loading information, glass sample slides for size resolved soiling impacts, and monitoring of rain, RH, and panel temperature for insight into meteorological impacts on soiling rates. Results indicate soiling reduces PV energy production by 0.37 ± 0.09% day^-1. The low cost digital microscope estimated soiling within ~1% of measured losses, confirming the feasibility of this low-cost alternative to expensive soiling stations.” ĐOẠN 2 (Section 2.2: Soiling measurements, p. 308): “Data for determination of soiling loss were obtained by a Campbell Scientific SMP100 soiling station. This soiling station used two, 50 Watt polycrystalline solar panels facing due south at a tilt angle of 23°. The reference panel was cleaned nightly with a microfiber cleaning towel and deionized (DI) water, while PM naturally deposited on the soiled panel.”	M. Valerino, M. Bergin, C. Ghoroi, A. Ratnaparkhi, and G. P. Smestad (2020) Sol. Energy, vol. 204, pp. 307–315. Link trực tiếp: DOI: 10.1016/j.solener.2020.03.118	 Solar Energy 204 (2020) 307–315 Contents lists available at ScienceDirect Solar Energy journal homepage: www.elsevier.com/locate/solener Low-cost solar PV soiling sensor validation and size resolved soiling impacts: A comprehensive field study in Western India Michael Valerino, Mike Bergin, Chinmay Ghoroi, Aniket Ratnaparkhi, Greg P. Smestad ARTICLE INFO ABSTRACT Deposition of particulate matter (PM) onto solar photovoltaic (PV) panels - known as soiling - has been estimated to reduce energy production by 10–40% in many regions of the world. Despite this, many key properties including soiling rates, PM source contributions, physical and optical properties of the deposited particles, and the impact of rain and relative humidity (RH) are not well understood. With this in mind we conducted a field study in Gandhinagar, India. Our approach combines soiling monitoring with a reference station and a low-cost digital microscopy system, sample collection for mass loading information, glass sample slides for size resolved soiling impacts, and monitoring of rain, RH, and panel temperature for insight into meteorological impacts on cleaning and soiling rates. Results indicate soiling reduces PV energy production by 0.37 ± 0.09% day^{-1}. The low cost (< 100$) digital microscope estimated soiling within ~1% of measured losses, confirming the feasibility of this low-cost alternative to expensive soiling stations. Deposited PM decreased energy production by 5.12 ± 0.55% per PM mass loading (g m^{-2}). Microscopy analyses of field samples revealed that > 90% of deposited mass loading is from particles > 5 μm in diameter, with > 50% of the soiling impacts estimated to be from particles < 5 μm. While heavy rain cleaned PV panels, light rains and high RH contributed to a 2x soiling rate and 5-10x PM deposition velocities as compared to dry periods. 1. Introduction In developing countries, solar photovoltaic (PV) power represented the largest renewable energy investment at 57.5 billion USD in 2016 (REN21, 2018). India has a projected goal of 100 GW (GW) of PV by 2022. Dust and particulate matter (PM) deposition on solar panels - Padmavathi and Daniel, 2013; Shukla et al., 2016; Sundaram et al., 2015; Sharma and Chandel, 2016). A key reason for the lack of reliable soiling information is the expensive cost of soiling monitoring equipment. There is a need for low-cost tools that can provide accurate insight into soiling effects. In addition, it would be very advantageous to combine soiling information with more detailed analysis of the de- *Chú thích: Trang 1 (p. 307), Tạp chí Solar Energy (Elsevier) của Michael Valerino et al. - Nghiên cứu đối chuẩn cảm biến quang học đo tổn thất bám bụi (Optical Sensor). Công trình dùng kính hiển vi kỹ thuật số phân tích lam kính mẫu vi mô; do tiêu chí (1) của Bảng 1 giới hạn ở camera thị giác máy tính giám sát bề mặt module PV toàn diện nên ghi nhận NR để phân biệt với hướng thị giác quang học vĩ mô.*

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				 are impacting PV energy production. In addition, there is not a clear agreement in the field on how meteorological variables, including rain and relative humidity (RH), impact both soiling rates and post-deposition processes like natural cleaning and cementation. With all of this in mind, we have developed a field-based approach that took place in Gandhinagar, Gujarat, India. We have combined traditional soiling monitoring using a Campbell Scientific Soiling Station with a low-cost digital microscope system which, to our knowledge, is a method that has never been validated. Soiling monitoring has been combined with two types of sample collection. Deposited PM has been collected from the surface of panels to determine mass loadings. To address the lack of size dependent soiling information, glass slide samples (on which PM naturally deposited), have been analyzed for the impact of particle size distribution on soiling. Finally, the influence of rain, RH, and ambient PM concentrations is considered. This comprehensive study looks to address gaps in regional soiling knowledge to allow energy producers to make more informed management decisions regarding mitigation of soiling (Tanesab et al., 2018; Baras et al., 2016; Abu-Naser, 2017; Dolan et al., 2015; Jones et al., 2016). **2. Material and methods** *2.1. Study location* This study took place on the campus of the India Institute of Technology Gandhinagar (IITGN) in Gandhinagar, the capital of Gujarat, India. On an unobstructed rooftop of a building on the campus, monitoring and sample collection equipment was set up, maintained and run from September 1st 2018 to January 9th 2019. This period is split into 6 3-week sampling periods referred to as A (September 1 - September 26), B (September 27 - October 17), C (October 18 - November 7), D (November 8 - November 30), E (November 31 - December 19) and F (December 20 - January 9). *2.2. Soiling measurements* Data for determination of soiling loss were obtained by a Campbell Scientific SMP100 soiling station. This soiling station used two, 50 Watt polycrystalline solar panels facing due south at a tilt angle of 23° (latitude of Gandhinagar). The reference panel was cleaned nightly with a microfiber cleaning towel and deionized (DI) water, while PM naturally deposited on the soiled panel. A data logger (Campbell CR-PVS1) recorded short-circuit current (I_{sc}) output and module temperature of both at a five minute interval. I_{sc} was then multiplied by the open circuit voltage (V_{oc}) of the panel and the recording time interval (300 s) to obtain an estimated energy output in Joules. It should be noted this method is not the same as the IEC standard for soiling (IEC 61724). Although the fill factor (maximum power over the product of I_{sc} and V_{oc}) will be influenced by soiling, we assume a constant value that will be canceled out in the soiling loss calculation. This is defined by Eq. (1): $$SL = \left(1 - \frac{E_{soiled}}{E_{reference}}\right) * 100\% \quad (1)$$ where SL is the soiling loss (%) for a given day, E_{soiled} is daily energy production of the soiled panel (J), and $E_{reference}$ is the daily energy production of the reference (cleaned) panel (J). It should be noted that temperature differences due to soiling are not accounted for in the being collected, including delaying collection until winds were low ($< 1\ m\ s^{-1}$) and using an umbrella to block any remaining breeze. This dust collection method is similar to that described by Javed et al. (2017). The panel was then cleaned with DI water and a microfiber cloth to remove any remaining particles. The mass of the deposited PM was measured with a Mettler Toledo AG 245 balance (with a precision of ± 0.2 mg), with PM mass values ranging from 40 to 700 mg (mg). *2.3. Meteorological data* Meteorological data were recorded at five-minute intervals using a Campbell Scientific CR1000 datalogger. Meteorological data were needed for assessment of natural cleaning by rain and to assess the impact of humidity on soiling rates. Attached instrumentation included temperature and relative humidity (Campbell Scientific EE181-L probe), and a rain gauge (Texas Electronics TR-525I). To assess the impact of ambient PM on soiling, an inexpensive PM monitor (Zheng et al., 2018) was deployed. The monitor uses a low cost light scattering sensor (Plantower PMS3003) along with custom hardware and software for data collection and calibration. The sensor is accurate to ~ 10% of ambient concentrations (Zheng et al., 2018). *2.4. Low cost digital microscope system* In order to fulfil the need for a low-cost system to estimate soiling, a digital microscope (Celestron Handheld Digital Microscope Pro 5MP) was used to capture daily images of deposited PM. The microscope had a low-iron glass slide (VWR Vistavision Unimark 75 × 25 × 1 mm) secured directly above the digital camera mounted on the camera stand, which was exposed to the air pointed due south at 23°tilt. Image analyses were done on images taken at 11PM when the microscope's internal LED lights illuminated particles against the dark night sky (see Supplemental Information for description of the system layout). Images taken with this microscope are ~2750 × 2070 μm. The setup was similar to that described by Figgis et al. (2016). The system was cleaned simultaneously to the Campbell Scientific station (every 3 weeks) with DI water and a microfiber cloth. Parameters described in subsequent sections with a 'dm' subscript will be used to indicate they were derived from this digital microscope method. *2.5. Optical microscope analyses* To determine size distribution and size dependent impacts of soiling, low iron, glass cover slips (VWR 22 × 22 mm No 1.5 – 0.16 to 0.19 mm in thickness) were used as sample slides for imaging deposited PM. These slides were secured to a custom 3D printed mount via a small rubber band. This mount and slide piece were then attached to another 3D printed bracket bolted to the side of a solar panel (See Supplementary Information for an image of the setup in the field). The glass slides were left out for a 3-week sampling period before being collected. A Zeiss Axio Observer microscope with a Hamamatsu Orca ER digital camera (1344 × 1024 pixels) was used to acquire micrographs of the field samples. Images to be used for determination of size distribution, mass loading estimations, and soiling loss estimations were taken with 20x and 100x objective lenses. The microscope eyepiece adds 10x magnification as well. Throughout this manuscript, '20x' and '100x' will be used to refer to the images taken through the 20x and 100x objective microscope lens. The dimension of the 20x images *Chú thích: Trang 2 (p. 308), Mục 2 - Phương pháp đo tổn thất công suất (Đo P: ✓), công trình không có cơ cấu làm sạch (NR), không phân vùng (NR) và không dùng PLC (NR).*
				   Fig. 1. The filled tan trace is the soiling measurement as recorded by the Campbell Scientific soiling station (SL). The square black points are the mass loading estimations acquired by the digital microscope (ML_{dm}). Vertical blue bars are rain amounts during events. For the two asterisked days (on September 2nd and September 21st) the corresponding processed image from the digital microscope is shown to the right where particles are black against a white background. *Chú thích: Trang 3 (p. 309), Hình 1 - Đồ thị thực nghiệm đối sánh mức suy giảm quang học và tổn thất năng lượng do bụi bám.*

TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc
[10]	Bảng 1 (Bảng đối chuẩn, Trang 2) & Mục 1 (Tổng quan giải pháp): “Luo et al. (2025) [10] (Robot cơ học + YOLOv8) (Thị giác: ✓ | Phân vùng: NR | Cơ cấu làm sạch: ✓ | ĐK công nghiệp: NR | Kiểm chứng: NR | Đo P: NR)” Mục 1 (Tổng quan giải pháp): “...cơ cấu làm sạch cơ học [10]–[12];...”	ĐOẠN 1 (Title & Abstract, Front. Robot. AI, vol. 12, p. 1606774, 2025): “A photovoltaic panel cleaning robot with a lightweight YOLO v8” “Abstract: Cleaning photovoltaic (PV) panels is essential for a PV station, as dirt or dust reduces the effective irradiation of solar energy and weakens the efficiency of converting solar energy into free electrons. The inconsistent cleaning efficacy and safety challenges (associated with summarized voltage and current) of manual methods pose significant hurdles for PV stations. Therefore, this paper develops an automated cleaning robot equipped with PV detection, path planning, and action control. First, a lightweight Mobile-ViT (Mobile Vision Transformer) model incorporating a Self-Attention mechanism was utilized to improve YOLOv8 (You Only Look Once v8), achieving an accuracy of 91.08% and a processing speed of 215 frames per second (fps). Second, the A* and Dynamic Window Approach (DWA) path planning algorithms were improved... The results show that the algorithm can successfully clean PV arrays autonomously, increasing the system's efficiency rate by 23% after implementation.”	J. Luo, G. Wang, Y. Lei, D. Wang, Y. Chen, and H. Zhang (2025) Front. Robot. AI, vol. 12, p. 1606774. Link trực tiếp: DOI: 10.3389/frobt.2025.1606774	 frontiers | Frontiers in Robotics and AI TYPE Original Research PUBLISHED 31 October 2025 DOI 10.3389/frobt.2025.1606774 Check for updates OPEN ACCESS EDITED BY Vladan Papić, University of Split, Croatia REVIEWED BY Josip Musić, University of Split, Croatia Zdenko Kovacic, University of Zagreb, Croatia *CORRESPONDENCE Hongzhou Zhang, 12010001@taru.edu.cn RECEIVED 06 April 2025 ACCEPTED 24 July 2025 PUBLISHED 31 October 2025 CITATION Luo J, Wang G, Lei Y, Wang D, Chen Y and Zhang H (2025) A photovoltaic panel cleaning robot with a lightweight YOLO v8. *Front. Robot. AI* 12:1606774. doi: 10.3389/frobt.2025.1606774 COPYRIGHT © 2025 Luo, Wang, Lei, Wang, Chen and Zhang. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms. A photovoltaic panel cleaning robot with a lightweight YOLO v8 Jidong Luo[1,2,3], Guoyi Wang[1,2,3], Yanjiao Lei[1,2,3], Dong Wang[1,2,3], Yayong Chen[3] and Hongzhou Zhang[1,2,3]* [1]College of Mechanical and Electrical Engineering, Tarim University, Alar, China, [2]Modern Agricultural Engineering Key Laboratory at Universities of Education Department of Xinjiang Uygur Autonomous Region, Alar, China, [3]Xinjiang Production and Construction Corps (XPCC) Key Laboratory of Utilization and Equipment of Special Agricultural and Forestry Products in Southern Xinjiang, Alar, China Cleaning PV (photovoltaic) panels is essential for a PV station, as dirt or dust reduces the effective irradiation of solar energy and weakens the efficiency of converting solar energy into free electrons. The inconsistent (cleaning efficacy) and unsafe (summarized voltage and current) manual method is a challenge for a PV station. Therefore, this paper develops a cleaning robot with PV detection, path planning, and action control. Firstly, a lightweight Mobile-ViT (Mobile Vision Transformer) model with a Self-Attention mechanism was used to improve YOLOv8 (You Only Look Once v8), resulting in an accuracy of 91.08% and a processing speed of 215 fps (frames per second). Secondly, an A* and a DWA (Dynamic Window Approach) path planning algorithm were improved. The simulation result shows that the time consumption decreased from 1.19 to 0.66 s and the Turn Number decreased from 23 to 10 p (places). Finally, the robot was evaluated and calibrated in both indoor and outdoor environments. The results showed that the algorithm can successfully clean PV arrays without manual control, with the rate increasing by 23% after its implementation. This study supports the maintenance of PV stations and serves as a reference for technical applications of deep learning, computer vision, and robot navigation. KEYWORDS cleaning robot, photovoltaic station management, lightweight YOLO v8, deep learning enhancement, path planning Highlights 1. YOLO v8 has been lightweighted with the Mobile-ViT for the cleaning robot. 2. The localizing and path planning algorithms were improved and validated in a computer environment. 3. Autonomous navigation and operation of robot prototypes in real-world environments. 1 Introduction Cleaning photovoltaic (PV) panels for a PV station (solar power station) is a Chú thích: Trang 1 (p. 1), Tạp chí Frontiers in Robotics and AI (Vol. 12, 2025, Article 1606774) của Jidong Luo et al. - Tiêu đề bài báo 'A Photovoltaic Panel Cleaning Robot with a Lightweight YOLO v8' và Tóm tắt xác nhận robot sử dụng mô hình Mobile-ViT tối ưu YOLOv8 phát hiện tấm pin và thuật toán A*/DWA lập kế hoạch đường đi.

TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc



FIGURE 16
Results of PV panel cleaning in real-world environments.

better than those of the other algorithms. On the contrary, the YOLO v5s model performs worse in all performance metrics, due to its use of a shallower feature extraction network. As a result, most performance evaluation metrics show lower levels. The results of this experiment demonstrate the superiority of the YOLO v8n-PP model in detecting the positional attitude of the PV panels.

To compare the recognition effect of different algorithm models on PV panel orientation more graphically, the optimal training weight file for each algorithm model is used to visualise and analyse the test set. The visualization results are shown in Figure 8. In this experiment, pictures with overlapping PV backgrounds and containing two types of orientations are selected as the objects of comparison. The comparative analysis reveals that the recognition accuracy of YOLO v8n surpasses that of the previous three algorithms. However, when dealing with PV panels positioned in overlapping backgrounds, the algorithm may appear to recognise PV targets that are heavily occluded, resulting in cluttered recognition target frames. It may also appear to recognize overlapping back-row PV panels. This phenomenon may cause serious interference with the robot, potentially preventing it from conducting the cleaning operation correctly. The YOLO v8n-PP further enhances the comprehensive performance of target detection and simultaneously enables accurate detection of the current PV panels to be cleaned

It fluctuates with the increase in the number of rounds in deep learning. It is only after the 100th round that the *P* stabilizes and remains above 90%. However, until the end of training, the curve still exhibits significant fluctuations, indicating that the *P* is not stable enough. In contrast, the YOLO v8n-PP algorithm has an initial *P* about 10% higher relative to YOLO v8n in the first 100 rounds of deep learning, and the *P* grows more smoothly as the number of deep learning rounds increases. Before the 100th round, the *P* has reached a steady state, which persists until the end of training, and the *P* curve remains stable, consistently higher than that of YOLO v8n.

Demonstrates the YOLO v8n and YOLO v8n-PP models in the training process of *mAP@0.5* visual comparison. According to the analysis of the experimental data, YOLO v8n's *mAP@0.5* remains volatile and has not yet reached a stable state at the end of training. Still, YOLO v8n-PP of *mAP@0.5* is larger than YOLO v8n and shows a smoother curve, which flattens out at the late stage of training. From the visualization results, the superiority of the YOLO v8n-PP model extends beyond numerical level advantages; the stability of its recognition ability is also crucial. The stability of its recognition ability is excellent in both the training and testing phases, consistently providing high-quality recognition results. This stability enables the model to remain efficient and accurate when managing complex tasks. Due to this, the YOLO v8n-PP model demonstrates excellent applicability when performing PV position recognition tasks on physical devices. Its stable recognition capability and excellent numerical performance enable the model to effectively cope with a variety of complex environments and conditions, thus providing great convenience in practical operations.

To verify the effectiveness of the improvement algorithm, three improvement YOLO v8n models were built for ablation-style experiments, including YOLO v8n-M, YOLO v8n-MPD, and YOLO v8n-PP. Among them, YOLO v8n-M is a lightweight model suitable for mobile devices, replacing Mobile-ViT as the Backbone. YOLO v8n-MPD is to replace class Bbox-Loss and class v8 detection loss in the detection HEAD module with the BBR loss function class LMPD-IoU-Loss. finally, YOLO v8n-PP is to replace class LMPD-IoU-Loss with class LMPD-IoU-Loss (this study model). Position recognition is a crucial visual task, and the original YOLO v8n has demonstrated strong reliability in this field. By analyzing the results of the ablation-style experiments in Table 2, the original YOLO v8n shows strong reliability in PV panel bit-position recognition. The initial deep learning using the unimproved YOLO v8n achieved 91.43%, 83.33%, 88.95%, 87%, and 126 fps for *P, R, mAP@0.5, F1*, and *FPS*, respectively. The excellent performance of these evaluated metrics highlights the potential of the YOLO v8n algorithm for PV bit-posture recognition. However, the algorithm still suffers from problems in the evaluation of detection metrics, such as low *F1*.

Chú thích: Trang 17 (p. 17), Hình 16 - Ảnh thực tế nguyên mẫu xe robot di động bánh xích tích hợp camera và chổi lau quét mặt tấm pin (Thị giác: ✓, Làm sạch: ✓).

TABLE 5 Contrasts with several similar studies.

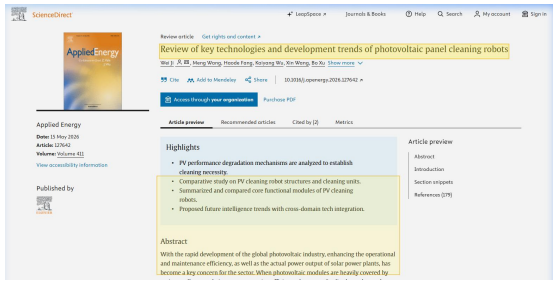
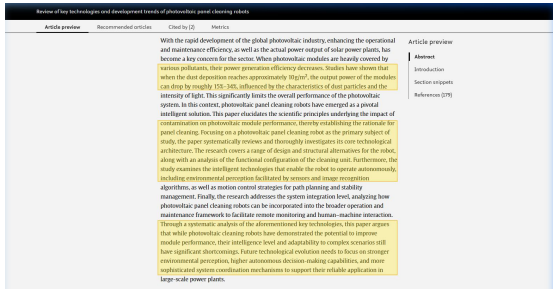
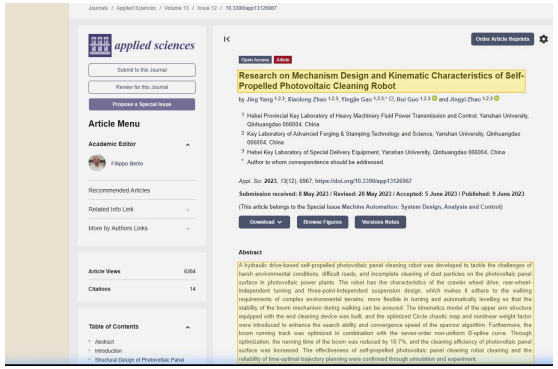
Method	Target detection method	Path planning algorithm	Cleaning method	PV panel condition
Ma method (Ma et al., 2022)	YOLO-PX	-	-	All
Prabhakaran method (Prabhakaran et al., 2022)	RMVDM	-	-	All
Grando robot (Grando et al., 2019)	-	-	Nozzles and water	With rail
Jin robot (Jin et al., 2024)	-	ACO algorithm	Both dry or with water	With rail
Yang robot (Yang et al., 2023)	-	-	Dry with brushes	Tilted and grounded
Riawan robot (Riawan et al., 2018)	-	-	Nozzles and water	Horizontal off-ground
Introduced robot	YOLO-v8n-PP	Location DWA	Nozzles and water	Tilted and grounded

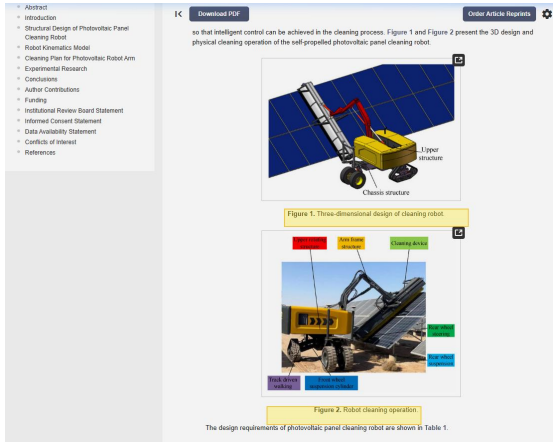
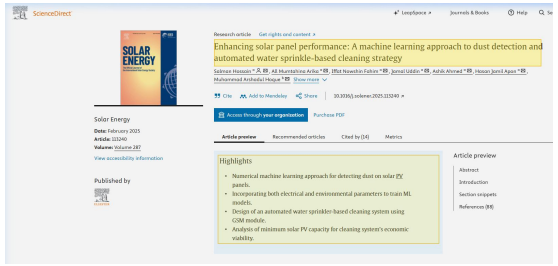
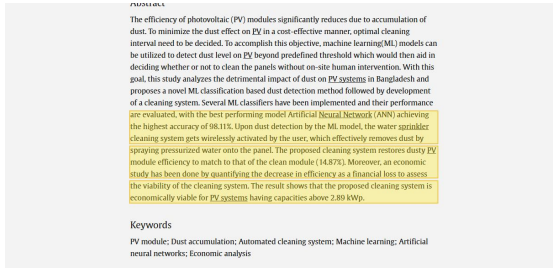
detection *FPS* of 239 fps. Compared to the unimproved YOLO v8n, the *P, R, mAP@*0.5, *F1*, *mAP@*0.5: 0.95, and *FPS* of YOLO v8n-M were raised by 1.52%, 8.9%, 3.97%, 5%, 4.14%, and 113 fps, and the *GFLOPs* was reduced by 2.4 t. It means that replacing Mobile-ViT for the network backbone can lead to a significant improvement in comprehensive network performance. After replacing the MPDIoU, the *P, R, mAP@*0.5, and *F1* of YOLO v8n-MPD were improved by 0.76%, 1.28%, 1.11%, and 1%, but the *GFLOPs* was also increased by 0.5 t. and its *mAP@*0.5:0.95, and the *FPS* decreased by 0.56% and

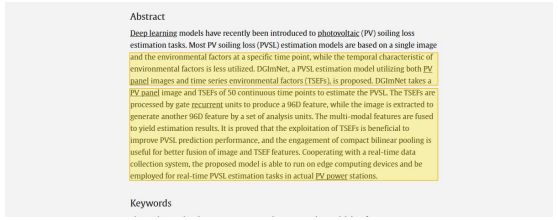
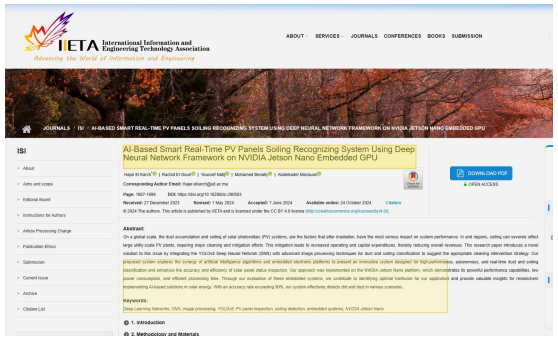
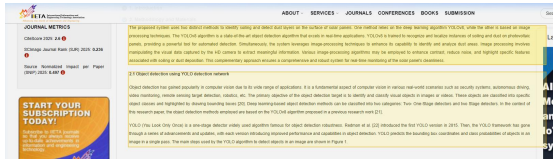
the experimental results of the two cases are compared. The simulation results are shown in Figure 10.

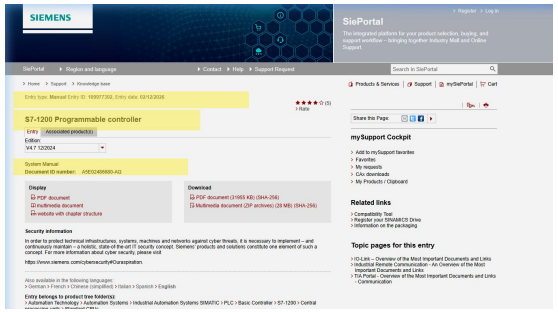
Table 3 shows the results of the simulation experiments are as follows: under the 30 × 30 raster map, the optimized algorithm with heuristic function increases the length of the travelling path by 6.9%, but shortens the search time by 73.61%; under the 60 × 60 raster map, the optimized algorithm increases the length of the travelling path by 14.29%, but shortens the search time by 79.3%. These experimental results demonstrate that the A* algorithm optimised by the

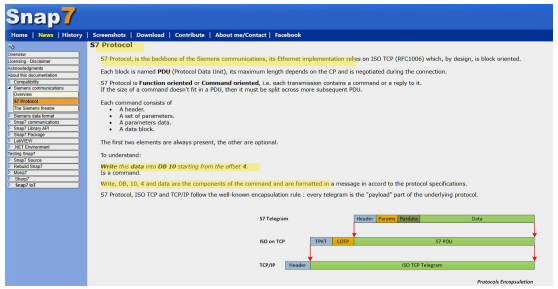
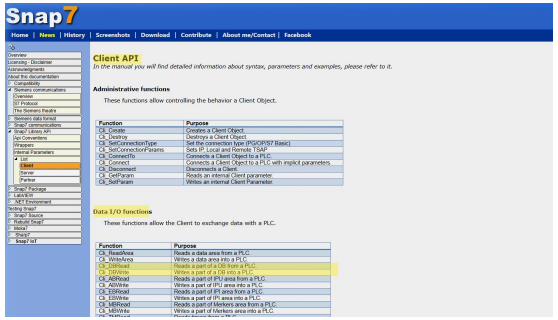
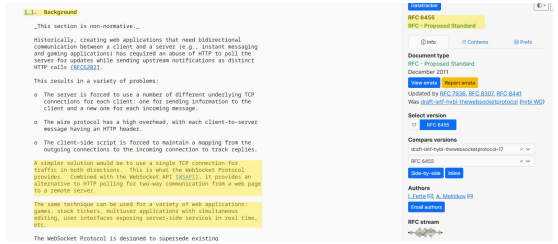
Chú thích: Trang 18 (p. 18), Bảng 5 - Đối chuẩn xác nhận robot nhận diện vùng bụi toàn khung để điều hướng; công trình không phân vùng 32 ô định lượng (NR), không dùng PLC công nghiệp (NR) và không đo công suất tại chỗ (NR).

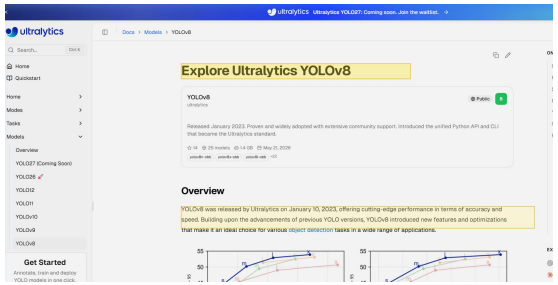
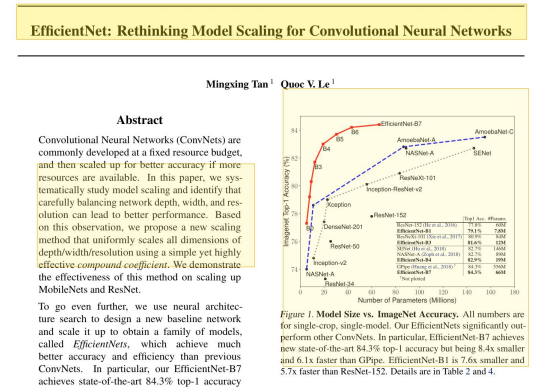
TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc
[11]	Mục 1 (Tổng quan giải pháp): "Các giải pháp tự động hiện nay chia thành bốn hướng chính: cảm biến quang học [9]; cơ cấu làm sạch cơ học [10]–[12]; học máy dự báo trên dữ liệu điện – môi trường [13]; và thị giác máy tính kết hợp học sâu [14], [15]."	*ĐOẠN 1 (Title & Highlights, ScienceDirect Elsevier, 2026):* *"Review of key technologies and development trends of photovoltaic panel cleaning robots"* *Highlights:* *• PV performance degradation mechanisms are analyzed to establish cleaning necessity.* *• Comparative study on PV cleaning robot structures and cleaning units.* *• Summarized and compared core functional modules of PV cleaning robots.* *• Proposed future intelligence trends with cross-domain tech integration.* *ĐOẠN 2 (Abstract, Applied Energy, vol. 411, p. 127642):* *"Studies have shown that when the dust deposition reaches approximately 10g/m2, the output power of the modules can drop by roughly 15%–34%... while photovoltaic cleaning robots have demonstrated the potential to improve module performance, their intelligence level and adaptability to complex scenarios still have significant shortcomings. Future technological evolution needs to focus on stronger environmental perception, higher autonomous decision-making capabilities, and more sophisticated system coordination mechanisms..."*	W. Ji, M. Wang, H. Fang, K. Wu, X. Wang, and B. Xu (2026) Appl. Energy, vol. 411, p. 127642. Link trực tiếp: DOI: 10.1016/j.apenergy.2026.127642	 Chú thích: Trang xuất bản ScienceDirect, Tạp chí Applied Energy (Vol. 411, 2026, Article 127642, Elsevier) của Wei Ji et al. - Tổng quan công nghệ và xu hướng phát triển của robot làm sạch pin PV.  Chú thích: Toàn văn Tóm tắt (Abstract) trên ScienceDirect minh chứng bụi bám làm suy giảm 15%–34% sản lượng điện và phân tích các cơ cấu làm sạch cơ học hiện đại.
[12]	Bảng 1 (Bảng đối chuẩn, Trang 2) & Mục 1 (Tổng quan giải pháp): "Yang et al. (2023) [12] Robot bánh xích tự hành (Thị giác: NR | Phân vùng: NR | Cơ cấu làm sạch: ✓ | ĐK công nghiệp: NR | Kiểm chứng: NR | Đo P: NR)" Mục 1 (Tổng quan giải pháp): "...cơ cấu làm sạch cơ học [10]–[12];..."	*ĐOẠN 1 (Title & Abstract, MDPI Applied Sciences, vol. 13, 6967, 2023):* *"Research on mechanism design and kinematic characteristics of self-propelled photovoltaic cleaning robot* *Abstract: A hydraulic drive-based self-propelled photovoltaic panel cleaning robot was developed... The robot has the characteristics of the crawler wheel drive, rear-wheel-independent turning and three-point-independent suspension design... The kinematics model of the upper arm structure equipped with the end cleaning device was built..."* *ĐOẠN 2 (Section 2 & Figure 1 & Figure 2, p. 3):* *"Figure 1. Three-dimensional design of cleaning robot... Figure 2. Robot cleaning operation. (Track driven walking, Front wheel suspension cylinder, Rear wheel steering, Rear wheel suspension)."*	J. Yang, X. Zhao, Y. Gao, R. Guo, and J. Zhao (2023) Appl. Sci., vol. 13, no. 12, p. 6967. Link trực tiếp: DOI: 10.3390/app13126967	 ***Chú thích:*** *Trang xuất bản MDPI, Tạp chí Applied Sciences (Vol. 13, Issue 12, Article 6967, 2023) của Jing Yang et al. - Thiết kế cơ cấu robot làm sạch pin tự hành dẫn động bánh xích.*

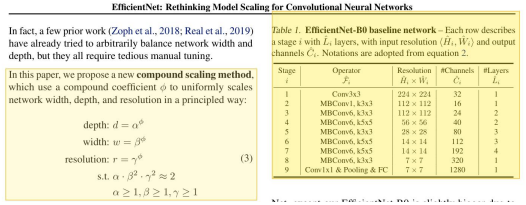
TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc
				 ***Chú thích:*** *Bản vẽ thiết kế 3D (Hình 1) và Ảnh chụp thực địa (Hình 2) trong bài báo của Yang et al. (2023) - Minh chứng nguyên mẫu robot bánh xích tự hành (Làm sạch: ✓); không thị giác (NR), không phân vùng (NR), không dùng PLC (NR) và không đo P (NR).*
[13]	Bảng 1 (Bảng đối chuẩn, Trang 2) & Mục 1 (Tổng quan giải pháp): "Hossain et al. (2025) [13] (Học máy dự báo + Phun nước) (Thị giác: NR | Phân vùng: NR | Cơ cấu làm sạch: ✓ | ĐK công nghiệp: NR | Kiểm chứng: NR | Đo P: ✓)" Mục 1 (Tổng quan giải pháp): "...học máy dự báo trên dữ liệu điện – môi trường [13];..."	*ĐOẠN 1 (Title & Highlights, ScienceDirect Elsevier, 2025):* *"Enhancing solar panel performance: A machine learning approach to dust detection and automated water sprinkle-based cleaning strategy"* *Highlights:* *• Numerical machine learning approach for detecting dust on solar PV panels.* *• Incorporating both electrical and environmental parameters to train ML models.* *• Design of an automated water sprinkler-based cleaning system using GSM module.* *ĐOẠN 2 (Abstract, Solar Energy, vol. 287, p. 113240):* *"Several ML classifiers have been implemented and their performance are evaluated, with the best performing model Artificial Neural Network (ANN) achieving the highest accuracy of 98.11%. Upon dust detection by the ML model, the water sprinkler cleaning system gets wirelessly activated... restores dusty PV module efficiency to match to that of the clean module (14.87%)."*	S. Hossain, A. M. Arika, I. N. Fahim, J. Uddin, A. Ahmed, H. J. Apon, and M. A. Hoque (2025) "Enhancing solar panel performance: A machine learning approach to dust detection and automated water sprinkle-based cleaning strategy," Sol. Energy, vol. 287, p. 113240. Link trực tiếp: DOI: 10.1016/j.solener.2025.113240	 *Chú thích: Trang xuất bản ScienceDirect, Tạp chí Solar Energy (Vol. 287, 2025, Article 113240) của Salman Hossain, All Mumtahina Arika et al. - Nghiên cứu hệ thống phun nước áp lực làm sạch pin PV dựa trên học máy (Phun nước: ✓).*  *Chú thích: Toàn văn Tóm tắt (Abstract) trên ScienceDirect - Mô hình ANN phân loại bụi trên dữ liệu điện và môi trường điều khiển phun nước tự động; không có thị giác máy chia lưới (NR), không dùng PLC (NR) và có đo khôi phục hiệu suất quang điện (Đo P: ✓).*

TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc
[14]	Bảng 1 (Bảng đối chuẩn, Trang 2) & Mục 1 (Tổng quan giải pháp): "Fang et al. (2024) [14] (Mạng nơ-ron DGImNet) (Thị giác: ✓ | Phân vùng chia lưới: ✓ | Cơ cấu làm sạch: NR | ĐK công nghiệp: NR | Kiểm chứng: NR | Đo P: NR)" Mục 1 (Tổng quan giải pháp): "...thị giác máy tính kết hợp học sâu [14], [15]."	*ĐOẠN 1 (Title & Highlights, ScienceDirect Elsevier, 2024):* *"DGImNet: A deep learning model for photovoltaic soiling loss estimation based on dust grid images"* *Highlights:* *• Build a model using images and time-series features to estimate PV soiling loss.* *• Proposed model is light-weighted and capable of running on edge computing devices.* *ĐOẠN 2 (Abstract, Applied Energy, vol. 376, Part B, p. 124335):* *"DGImNet, a PVSL estimation model utilizing both PV panel images and time series environmental factors (TSEFs), is proposed. DGImNet takes a PV panel image and TSEFs of 50 continuous time points to estimate the PVSL... is able to run on edge computing devices and be employed for real-time PVSL estimation tasks in actual PV power stations."*	M. Fang, W. Qian, T. Qian, Q. Bao, H. Zhang, and X. Qiu (2024) Appl. Energy, vol. 376, Part B, p. 124335. Link trực tiếp: DOI: 10.1016/j.apenergy.2024.124335	 ***Chú thích:*** *Trang xuất bản ScienceDirect, Tạp chí Applied Energy (Vol. 376, Part B, 2024, Article 124335, Elsevier) của Mingyu Fang et al. - Mô hình học sâu DGImNet phân tích ảnh chia lưới bụi để ước lượng tổn hao bám bụi (Thị giác: ✓, Phân vùng: ✓).*  ***Chú thích:*** *Toàn văn Tóm tắt (Abstract) trên ScienceDirect - Xác nhận mô hình DGImNet triển khai trên thiết bị tính toán biên; không tích hợp cơ cấu làm sạch (NR), không điều khiển bằng PLC (NR) và không đo công suất trực tiếp (NR).*
[15]	Bảng 1 (Bảng đối chuẩn, Trang 2) & Mục 1 (Tổng quan giải pháp): "El Karch et al. (2024) [15] (YOLOv8 nhúng Jetson) (Thị giác: ✓ | Phân vùng chia lưới: NR | Cơ cấu làm sạch: NR | ĐK công nghiệp: NR | Kiểm chứng: NR | Đo P: NR)" Mục 1 (Tổng quan giải pháp): "...thị giác máy tính kết hợp học sâu [14], [15]."	*ĐOẠN 1 (Title & Abstract, ISI Journal, vol. 29, no. 5, 2024):* *"AI-Based Smart Real-Time PV Panels Soiling Recognizing System Using Deep Neural Network Framework on NVIDIA Jetson Nano Embedded GPU* *Abstract: This research paper introduces a novel solution by integrating the YOLOv8 Deep Neural Network with advanced image processing techniques for dust and soiling classification... Our approach was implemented on the NVIDIA Jetson Nano platform... With an accuracy rate exceeding 90%, our system effectively detects dirt and dust in various scenarios."* *ĐOẠN 2 (Section 2 & Section 2.1, pp. 1688-1689):* *"2. Methodology and Materials: The proposed system uses two distinct methods to identify soiling and detect dust layers... One method relies on the*	H. El Karch, R. El Gouri, Y. Natij, M. Benally, and A. Mezouari (2024) Ingénierie des Systèmes d'Information, vol. 29, no. 5, pp. 1687–1699. Link trực tiếp: DOI: 10.18280/isi.290503	 *Chú thích: Trang xuất bản IIETA, Tạp chí Ingénierie des Systèmes d'Information (Vol. 29, No. 5, 2024, pp. 1687–1699) của Hajar El Karch et al. - Ứng dụng YOLOv8 trên GPU nhúng NVIDIA Jetson Nano nhận diện phân loại bám bẩn với độ chính xác >90% (Thị giác: ✓ | Phân vùng chia lưới: NR).* 

TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc
		deep learning algorithm YOLOv8...”		*Chú thích: Phương pháp luận và Kiến trúc mạng YOLOv8 (Mục 2 & 2.1) - Định vị vùng bám bản toàn khung bằng hộp bao (bounding box); công trình thuần giám sát AI biên, không phân vùng chia lưới cố định (NR), không có cơ cấu làm sạch (NR), không PLC (NR), không đo công suất (NR).*
[16]	Mục 2.1 (Cấu hình phần cứng) & Mục 2.5 (Điều khiển chu trình PLC): “...Cơ cấu chấp hành gồm ray trượt 670 mm, chổi gạt hai chiều, bơm áp lực 0,5 HP, PLC Siemens S7-1200 [16], camera giám sát 1080p...” (Mục 2.1) “PLC S7-1200 CPU 1214C [16] điều khiển chu trình theo máy trạng thái hữu hạn.” (Mục 2.5)	*ĐOẠN 1 (Siemens SiePortal SIOS - S7-1200 Programmable Controller System Manual, Entry ID: 109977302):* *“SIMATIC S7-1200 Programmable controller System manual. Entry type: Manual, Entry ID: 109977302, Edition: 2024, Document ID: A5E02486680-AQ. Automation Technology > Automation Systems > SIMATIC > PLC > S7-1200 > Central processing units > CPU 1214C.”* *ĐOẠN 2 (Technical Specifications > CPU 1214C > Table: Performance):* *“CPU 1214C DC/DC/DC | Article number: 6ES7214-1AG40-0XB0* *Type of instruction: Boolean | Execution speed: 0.08 µs/instruction* *Type of instruction: Move Word | Execution speed: 0.137 µs/instruction* *Type of instruction: Real Math | Execution speed: 1.48 µs/instruction* *Note: Performance times are for the fastest instructions in this category and error-free programs.”*	Siemens AG (2024) SIMATIC S7-1200 Programmable Controller System Manual Siemens Industry Online Support (SIOS), Entry ID: 109977302. Link trực tiếp: Cổng thông tin kỹ thuật SIOS Siemens	 *Chú thích: Trang danh mục kỹ thuật chính thức Siemens SiePortal SIOS - Sổ tay hệ thống SIMATIC S7-1200 Programmable Controller System Manual (Entry ID: 109977302, Edition 12/2024) - Xác nhận nền tảng phần cứng CPU 1214C, tốc độ thực thi chỉ lệnh logic (0,08 µs/lệnh) và giao diện kết nối Ethernet công nghiệp. Các ngưỡng công nghệ (Ctrigger, Cverify) và thuật toán máy trạng thái FSM là giải pháp thiết kế riêng của nhóm nghiên cứu, không thuộc quy định của sổ tay phần cứng.* Table: Performance Type of instruction / Execution speed / Direct addressing (I, Q and M) / DB accesses Boolean — 0.08 µs/instruction Move — Move_Bool — 0.3 µs/instruction — 1.17 µs/instruction Move_Word — 0.137 µs/instruction — 1.0 µs/instruction Move_Real — 0.72 µs/instruction — 1.0 µs/instruction Real Math — Add Real — 1.48 µs/instruction — 1.78 µs/instruction Note: Many variables affect measured times. The above performance times are for the fastest instructions in this category and error-free programs. *Chú thích: Bảng thông số hiệu năng xử lý (Table: Performance) của CPU 1214C trích từ Sổ tay kỹ thuật Siemens - Tốc độ thực thi chỉ lệnh logic Boolean đạt 0,08 µs/lệnh, là nền tảng phần cứng bảo đảm CPU thực thi nhanh chóng các tác vụ điều khiển logic.*

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[17]	Mục 1 (Mở đầu), Mục 2.1 (Kiến trúc hệ thống) & Mục 2.5 (Điều khiển PLC): "...lệnh điều khiển được gửi tới PLC qua thư viện Snap7 [17] (cổng 102) để thực thi chu trình gạt tương ứng..." (Mục 2.1) "Máy tính biên gửi cờ qua Snap7 [17] (cổng 102): DB1.DBX0.0 (chu trình ướt xử lý bụi 30 s), DB1.DBX0.1 (chu trình khô quét lá 24 s)..." (Mục 2.5)	*ĐOẠN 1 (Snap7 Documentation - Siemens communications > S7 Protocol & Protocols Encapsulation):* *"S7 Protocol is the backbone of Siemens communications, its Ethernet implementation relies on ISO TCP (RFC1006) which, by design, is block oriented. Each block is named PDU (Protocol Data Unit)... S7 Protocol is Function oriented, every telegram is the 'payload' part of the underlying protocol. (Protocols Encapsulation: S7 Telegram -> ISO on TCP -> TCP/IP over Port 102)."* *ĐOẠN 2 (Snap7 Library API > Client API > Data I/O functions):* *"Data I/O functions allow the Client to exchange data with a PLC:* *Cli_DBRead: Reads a part of a DB from a PLC.* *Cli_DBWrite: Writes a part of a DB into a PLC."*	D. Nardella (2021) Snap7: An Open Source Multi-platform Ethernet S7 Communication Suite SourceForge Project Documentation, Version 1.4.2. Link trực tiếp: Tài liệu kỹ thuật Snap7 SourceForge	 *Chú thích: Tài liệu kỹ thuật chính thức Snap7 Documentation (snap7.sourceforge.net) - Đặc tả thư viện Snap7 giao tiếp giao thức S7 Protocol trên nền tảng ISO on TCP (RFC 1006, cổng 102) và sơ đồ đóng gói bức điện S7 Telegram.*  *Chú thích: Bảng chỉ mục hàm Client API (Mục Data I/O functions) - Hai hàm Cli_DBRead và Cli_DBWrite của thư viện Snap7 cho phép đọc ghi dữ liệu trực tiếp với khối Data Block DB1 của PLC S7-1200 qua kết nối Ethernet TCP/IP.*
[18]	Mục 2.1 (Kiến trúc hệ thống) & Mục 3.5 (Giám sát SCADA/IoT): "Dữ liệu vận hành được lưu trữ SQLite và đồng bộ lên nền tảng SCADA/IoT qua giao thức WebSocket [18]." (Mục 2.1) "Web SCADA và ứng dụng React Native (Hình 7) kết nối máy tính biên qua WebSocket [18], hiển thị điện áp, dòng điện, công suất tức thời và trạng thái làm sạch." (Mục 3.5)	*ĐOẠN 1 (IETF RFC 6455 - Standards Track, December 2011, Title & Abstract, p. 1):* *"Internet Engineering Task Force (IETF) I. Fette, Google, Inc.* *Request for Comments: 6455 A. Melnikov, Isode Ltd.* *Category: Standards Track December 2011* *The WebSocket Protocol* *Abstract: The WebSocket Protocol enables two-way communication between a client running untrusted code in a controlled environment to a remote host that has opted-in to communications from that code... The protocol consists of an opening handshake followed by basic message framing, layered over TCP."* *ĐOẠN 2 (Section 1.1 Background, p. 2):* *"A simpler solution would be to use a single TCP connection for traffic in both directions. This is what the WebSocket Protocol provides. Combined with the WebSocket API, it provides an alternative to HTTP polling for two-way communication from a web page to a remote server... suitable for real-time user interfaces."*	I. Fette and A. Melnikov (2011) "The WebSocket Protocol", Internet Engineering Task Force (IETF), RFC 6455. Link trực tiếp: DOI: 10.17487/RFC6455 (IETF Standards Track)	 ***Chú thích:*** *Tiêu chuẩn chính thức RFC 6455 trên cổng IETF Datatracker - Tiêu đề 'The WebSocket Protocol', các tác giả I. Fette (Google) và A. Melnikov (Isode), định nghĩa giao thức truyền thông hai chiều thời gian thực.* 

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				Chú thích: *Phân mục 1.1 (Background) của RFC 6455 - Minh chứng nguyên lý sử dụng một kết nối TCP duy nhất cho truyền thông hai chiều thời gian thực phục vụ giao diện người dùng Web SCADA / ứng dụng di động.*
[19]	Mục 2.4 (Kiến trúc mô hình): "Chuỗi AI gồm YOLOv8n-Seg [19] (6,8 MB, 100 chu kỳ) định vị viền và nắn phối cảnh về khung 896 × 1792 px để tạo 32 ô 224 × 224 px (Hình 3)..."	ĐOẠN 1 (Ultralytics YOLOv8 Documentation - Overview & Features, 2023): "Explore Ultralytics YOLOv8: Released January 2023. Proven and widely adopted with extensive community support. Introduced the unified Python API and CLI that became the Ultralytics standard. Building upon previous YOLO versions, YOLOv8 introduced new features and optimizations that make it an ideal choice for object detection and instance segmentation tasks." ĐOẠN 2 (Official Performance Metrics Table - Instance Segmentation COCO): "Model: YOLOv8n-seg | size (pixels): 640 | mAP_mask (50-95): 30.5 | Speed CPU ONNX (ms): 96.1 | params (M): 3.4 | FLOPs (B): 12.0"	G. Jocher, A. Chaurasia, and J. Qiu (2023) "Ultralytics YOLOv8," Version 8.0.0. Link trực tiếp: Kho mã nguồn Ultralytics GitHub	 *Chú thích: Cổng tài liệu kỹ thuật chính thức Ultralytics YOLOv8 Docs (Phiên bản v8.0.0, Ultralytics Inc., 2023) - Tổng quan kiến trúc mạng YOLOv8 hỗ trợ phát hiện đối tượng và phân đoạn cá thể (instance segmentation).*  *Chú thích: Bảng thông số hiệu năng phân đoạn cá thể (Segmentation COCO) trên tài liệu chính thức Ultralytics - Mô hình YOLOv8n-seg có kiến trúc thu nhỏ với 3,4 triệu tham số (3.4M params), làm cơ sở cho mô hình phân đoạn viền pin của đề tài.*
[20]	Mục 2.4 (Huấn luyện mô hình): "...EfficientNet-B0 [20] huấn luyện 15 chu kỳ (Adam, LR = 0,0001; TFLite FP16, 10,6 MB; seed = 0) (Bảng 3)."	*ĐOẠN 1 (Title, Abstract & Figure 1, ICML 2019 / PMLR 97):* *"EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks* *Mingxing Tan, Quoc V. Le (Google Brain)* *Abstract: In this paper, we systematically study model scaling and identify that carefully balancing network depth, width, and resolution can lead to better performance. Based on this observation, we propose a new scaling method that uniformly scales all dimensions using a simple yet highly effective compound coefficient...* *Figure 1: EfficientNet-B0 achieves 76.3% top-1 accuracy with only 5.3M parameters, significantly outperforming conventional ConvNets while being much smaller and faster."* *ĐOẠN 2 (Table 1: EfficientNet-B0 Baseline Network, p. 6109):* *"Table 1. EfficientNet-B0 baseline network (MBConv blocks, 224x224 input resolution, 5.3M parameters)."*	M. Tan and Q. V. Le (2019) "EfficientNet: Rethinking model scaling for convolutional neural networks," in Proc. ICML 2019, PMLR 97, pp. 6105–6114. Link trực tiếp: Bản thảo gốc ICML 2019 / arXiv:1905.11946	 Figure 1. **Model Size vs. ImageNet Accuracy.** All numbers are for single-crop, single-model. Our EfficientNets significantly outperform other ConvNets. In particular, EfficientNet-B7 achieves new state-of-the-art 84.3% top-1 accuracy but being 8.4x smaller and 6.1x faster than GPipe. EfficientNet-B1 is 7.6x smaller and 5.7x faster than ResNet-152. Details are in Table 2 and 4. ***Chú thích:*** *Trang 1 kỷ yếu hội nghị quốc tế ICML 2019 (PMLR 97) của Mingxing Tan và Quoc V. Le (Google Brain) - Giới thiệu kiến trúc EfficientNet tối ưu hóa tỷ lệ mạng nơ-ron, mô hình EfficientNet-B0 đạt độ chính xác cao chỉ với 5,3 triệu tham số (5.3M params).*

TT	Vị trí & Luận điểm trích dẫn trong bài báo TNU	Đoạn văn nguyên văn đối chiếu từ tài liệu gốc	Nguồn tài liệu xuất bản & Định danh số	Hình ảnh minh chứng thực tế từ tài liệu gốc
				

EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks

In fact, a few prior work (Zoph et al., 2018; Real et al., 2019) have already tried to arbitrarily balance network width and depth, but they all require tedious manual tuning.

In this paper, we propose a new **compound scaling method**, which use a compound coefficient ϕ to uniformly scales network width, depth, and resolution in a principled way:

$$\begin{aligned} \text{depth: } & d = \alpha^\phi \\ \text{width: } & w = \beta^\phi \\ \text{resolution: } & r = \gamma^\phi \\ \text{s.t. } & \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \\ & \alpha \geq 1, \beta \geq 1, \gamma \geq 1 \end{aligned} \tag{3}$$

Table 1. **EfficientNet-B0 baseline network** – Each row describes a stage i with $\hat{L}_i$ layers, with input resolution $\langle \hat{H}_i, \hat{W}_i \rangle$ and output channels $\hat{C}_i$. Notations are adopted from equation 2.

Stage i	Operator $\hat{\mathcal{F}}_i$	Resolution $\hat{H}_i \times \hat{W}_i$	#Channels $\hat{C}_i$	#Layers $\hat{L}_i$
1	Conv3x3	224×224	32	1
2	MBConv1, k3x3	112×112	16	1
3	MBConv6, k3x3	112×112	24	2
4	MBConv6, k5x5	56×56	40	2
5	MBConv6, k3x3	28×28	80	3
6	MBConv6, k5x5	14×14	112	3
7	MBConv6, k5x5	14×14	192	4
8	MBConv6, k3x3	7×7	320	1
9	Conv1x1 & Pooling & FC	7×7	1280	1

Chú thích: *Công thức mở rộng tích hợp Compound Scaling (Equation 3) và Bảng kiến trúc mạng cơ sở EfficientNet-B0 (Table 1) - Minh chứng cấu trúc khối MBConv và kích thước đầu vào 224×224 px, làm cơ sở lý thuyết cho việc phân loại định lượng mức độ bám bẩn Cdirty trên 32 ô lưới.*